MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6AK6

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, miniature, power

© Outline --- 5-2 (EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	g3	h	h	a	g2	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ik	Pp	Pg2	Ehk	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	mAdc	W	W	v	°C	ft
Maximum:	6.9	300	---	275	21	3.0	0.83	180	140	(See note 1)
Minimum:	5.7	---	---	---	---	---	---	---	---	
TEST CONDITIONS:	6.3	180	-9	180	---	---	---	---	---	---

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 202640-15T12:50Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Eb = 300 V; Ec1 = -25 V; Ec2 = 300 V (see note 2)	0.65	II	Ic1	0	-1.5	µAdc
1256	Electrode current (anode)		0.65	II	Ib	11	19	mAdc
1341	Power output (1)	Esig = 6.35 Vac; Rp = 10,000 ohms	0.65	II	Po	900	---	mW
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 2,000 ohms	---	---	Ep	---	500	mVac
(C) 1336	Heater-cathode leakage		---	---	Ihk	---	20	µAdc
1301	Heater current		---	---	If	138	162	mA
1256	Electrode current (screen)		---	---	Ic2	1	4	mAdc
1306	Transconductance		---	---	Sm	1,900	2,600	µmhos
1341	Power output (2)	Ef = 5.5 V; Esig = 6.35 Vac; Rp = 10,000 ohms	---	---	Po	800	---	mW
1211	Insulation of electrodes		---	---	---	---	---	---
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A; Eb = 300 Vdc; Ec1 = -25 Vdc; Ec2 = 300 Vdc; Ehk = +180 V; Rg1 = 0.1 Meg (min)	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours)	Transconductance or Power output (1)	--- ---	--- ---	Sm Po	1,520 600	--- ---	µmhos mW

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:50Z
Check the source to verify that this is the current version before use.

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, MU
Navy -
Air Force - 11, 70, 85
DSA - ES

User activities:
Army - ME, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2401-50)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:50Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:50Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:50Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:50Z
Check the source to verify that this is the current version before use.